NLP Based Patient Record Summarization for Clinical Decision

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***Abstract*—As EMS creates and stores more unstructured clinical text in their electronic health records (EHRs), manual review becomes inefficient and time consuming. Natural Language Processing (NLP) offers a method that can be used to convert this information into a summarised, structured summary.**

In this paper, a hybrid system is proposed to capture key information from clinical texts using the extractive techniques of TF-IDF and generate coherent and context-aware text summaries using a fine-tuned BART model. A comparison with baseline approaches is conducted, such as the extractive TF-IDF, and the zero-shot pre-trained BART. Multiple benchmarks, such as ROUGE, BERTScore, METEOR and disease-specific metrics like medical entity preservation and clinical concept overlap.

The fine-tuned model-generated results indicate better summary quality than the baselines, both in accuracy and clinical relevance. The outputs generated are structured and easy to interpret and better suited to the use in clinical workflows. In general, the strategy has rendered it easier to gain access to information, decreases documentation requirements, and enhances clinical decision-making processes.

***Index Terms*—Natural Language Processing, Clinical Text Summarization, Electronic Health Records, BART, Medical NLP, Abstractive Summarization, Clinical Decision Support**

I. Introduction

As for electronic health, the continuous and recent use of electronic health (eHealth) were analyzed. Since the introduction of records (EHRs), there has has been a surge in the amount of clinical data. Information generated in the form of text data such as discharge summaries, Physician notes/diagnostic reports. While a vital source of biomedical knowledge, these records are often long and un-structured, making the task of healthcare professional fairly difficult in interpreting them swiftly. This often causes delays in the process of clinical visits and records may be reviewed for some significant period of time, potentially compromising efficiency and accuracy in locating the relevant information in records.

Extractive methods, e.g., TF-IDF, rule-based methods, etc., are the major methods used with traditional summarization techniques to summarize clinical text data. These methods detect important sentences by counting the frequency of certain sentences or by forcing them according to certain rules, but they can't create new documents. This means that summaries generated are short and may be disjointed and repetitive. They also have difficulties linking information from various passages in the text in a meaningful way.

Later research has been centred around Transformers, which have worked well on many NLP tasks. BART can be used to model text-to-text relationships, and to create more natural text summaries. Models developed from general domain data on the other hand, might not match the performance of data from a clinical domain without further adaptation due to divers differences in medical language being more specialised and often structured differently.

This study is based on the development of a structured summarization approach via fine tuning of BART model. This is a model that is able to generate a summary in a format similar to that of a JSON, covering various key aspects of a clinical picture like symptoms, diagnosis and treatment. The method is tested using both traditional NLPs and domain-specific measures for a more comprehensive picture of quality of and utility for summaries.

A. Objectives

This study aims to build and test a model that will be able to create correct and organized abstractions of unstructured clinical notes.

- To develop a fine-tuned BART model using a clinical text corpus architecture.
- To examine its performance with similar models and/or traditional methods such as TF-IDF and zero-shot BART.
- To use a variety of metrics to assess the performance of the generated summaries, including ROUGE, BERTScore, and domain-specific metrics like METEOR.
- To evaluate entity preservation and overlap between medical entities and clinical concepts.
- To understand how the quality and accuracy of the fine-tuned weights from earlier domain-specific training relate to fine-tuning using another domain.
- To evaluate the usefulness and accuracy of clinical summaries.

II. Literature Survey

Large language model (LLM) research. The advancements in Large Language Model (LLM) research have had an impact on medical text summarization. Language Models (LM) and Transformer-based models. Van Veen et al. [4] evidenced that AI models empowered to do a particular medical assignment can generate more accurate summaries or, in certain instances, generate documents that are near to the human-written report.

Their study also highlighted, however, issues like "hallucination" and "inconsistency," which are still significant problems in clinical applications [5].

Gupta et al. [6] focused on enhancing factual accuracy with the help of Retrieval-Augmented Generation (RAG), which incorporates external medical knowledge during the summarization process. This will help to increase reliability, Needs good retrieval systems and programme plans, particularly with regard to unstructured clinical data. Hartman et al. [7] investigated on transformer based encoder– The GenBank is utilized to generate models of the decoder type as BERT and BART. Models of the decoder, like BERT and BART models are used on GenBank. discharge summaries. Their research found that the models obtained from their work can learn clinic relevant knowledge without correct It is possible that he/she will have to generate summaries at times.

Furthermore, Schneider, et al. [8] wrote about an application of LLMs for summarizing texts come in various categories.There are different types of LLMs for summarizing texts. structured clinical data. They succeeded with their work, so that improvements were made. up in terms of scalability and data storage actually hinder productivity.But issues with scalability and data storage reduce productivity. Also observed were the frequency of the various species and their availability. Like the said survey reported by Ghosh et al. [11] Transformer based methods have some problems like hallucination, inconsistent assessment methods, and inconsistencies. in assessment.

Factual consistency was enhanced by data augmentation methods and domain adaptation methods used by Zhang et al. [9]. They validated their study with a controller set for domainspecific fine-tuning can improve the summarization performance, but also boosts the compute needs and for tight selection of components' specifications and sizes. datasets. There are only a few methods that exist that use the end-to-end model, but at a large scale. While a good summarization performance is made by these models, the text generated from these models is not controllable. and summary structure.Many studies also rely heavily on lexical overlap metrics such as ROUGE for evaluation, which may not fully capture semantic quality, despite showing high test–retest reliability and clinical applicability.

Therefore, the approach used in this work combines TF-IDF-based feature extraction with a fine-tuned BART model for generating structured summaries. This combined approach aims to provide better control over the generated information while maintaining clinical relevance and consistency in the final outputs.

III. Dataset

A. Dataset Description

The *AGBonnet/augmented-clinical-notes* (ACN) is utilized from the HuggingFace Datasets Hub as used in this study. The data set comprises of clinical notes associated with structured clinical summary text extracted from clinical case reports that is appropriate for supervised clinical summarization.

TABLE I
Dataset Statistics for AGBonnet/augmented-clinical-notes

Attribute	Value
Total samples	30,000
Samples used	5,000
Training set	3,500
Validation set	750
Test set	750
Input	Clinical note
Output	Structured summary
Median note length	∼220 words
Median summary length	∼180 words

The samples are accompanied by natural language clinical notes, together with the summary in the JSON-style, that contains important clinical fields, like the patient's health or details of symptoms, diagnosis and treatment. The length of the notes is different and address several areas of medicine such as cardiology, oncology and neuroscience.

B. Dataset Statistics

Total sample size is ∼30,000. Random sampling with seed = 42 was used to select a subset of 5000 data points, to ensure that there is enough diversity to allow computation while keeping it manageable. The data was split into training (70%), validation (15%), and test (15%) sets.

C. Dataset Characteristics

The data set has some difficulties. Medical clinical notes have high density of medical information. Abbreviations and terminology, which may not be easy on general-domain models. The source text isn't structured and is followed by target summaries that are presented in a structured manner. Additionally, a lot of fields are present in the structured summaries, so output length is important to avoid truncating important information. The data set also includes categorical variability spanning multiple medical specialties including diagnoses and treatments.

IV. Preprocessing

The dataset was processed through several preprocessing steps to improve data consistency and prepare the clinical text for training with the BART model.

A. Data Filtering

The data set was pruned to only include the relevant data fields. When they were cleaned, the samples with missing values were excluded and the exceptionally short samples with text post cleaning were excluded. It prevents unstable training which occurs when it fails to have valid or complete pairs of input/output.

B. Text Normalization

The text was uniformed to eliminate the noise and derive uniformity. This involved stripping out the unwanted formatting and some irregular whitespace, as well as normalizing the spacing. This is especially critical for clinical text, which

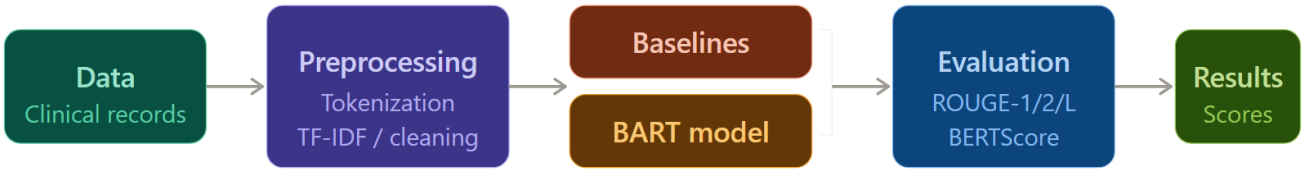


Fig. 1. Clinical text summarization pipeline.

may have inconsistent format issues from data entry and/or copy/paste errors.

C. Sampling and Data Split

Of the data cleaned, 5000 samples were obtained by random sampling with a fixed seed (42) for reproducible results. The data was then split to create a training set (70%), validation set (15%) and test set (15%).

D. Tokenization

The tokenizer used to tokenize the sentence is BART tokenizer (facebook/bart-large-cnn). 512 input sequence tokens could be maximum, 384 output summary tokens could be maximum. By adding more length to the target length property, important fields won't be truncated, keeping the full structurally formed summaries. Padding was applied to make batch sizes match, and batched processing was used for all splits for efficiency.

V. Methodology

A. Overall Pipeline

The proposed system follows a three-stage pipeline consisting of (i) data preparation and preprocessing, (ii) model training and summarization, and (iii) multi-metric evaluation. The overall pipeline is designed to support reproducible experimentation and end-to-end execution on GPU-based infrastructure.

B. Pipeline Overview

Figure 1 shows the complete workflow of the proposed system. The process begins with dataset ingestion, followed by preprocessing steps such as cleaning, sampling, and tokenization. The resulting data is then utilized as baseline and fine-tuned model evaluation. Lastly, the created summaries are evaluated with several evaluation criterion to evaluate quality of summarization and relevance to the clinical situation.

C. Baseline Models

We compare with two baseline approaches.

TF-IDF Extractive Model: The clinical note is divided into sentences, and the importance for each sentence is calculated using TF-IDF. . Cosine similarity is applied with the document representation, and the highest-ranked sentences are selected to form the summary. This method helps maintain factual consistency, but the generated summaries are often less coherent and lack abstraction.

Zero-shot BART: A pre-trained *facebook/bart-large-cnn* model is used without fine-tuning. This baseline is used to evaluate how a general-domain summarization model performs on clinical text without any domain-specific adaptation.

D. Proposed Model

The proposed model is an improved version of the facebook/bart-large-cnn transformer. This model is trained with clinical data via a sequence-to-sequence learning method. It was able to extract an overview and section from unstructured notes, create coherent and organized summaries from unstructured notes, and summarize texts. Inference decodes with a beam search algorithm with length constraints to ensure fluency and avoid repetition. This helps generate more detailed and accurate abstracts.

E. Evaluation Protocol

All models are tested using a held-out test set to ensure fair comparison. A large number of metrics are used to measure performance such as ROUGE, BERTScore, METEOR, and domain-specific metrics such as medical entity preservation and clinical concept overlap. This multi-metric setup provides a more accurate evaluation of both linguistic competence and clinical accuracy.

VI. Models and Training Configuration

A. Model Architecture

The proposed model architecture closely follows the *facebook/bart-large-cnn* transformer-based sequence-to-sequence network for abstractive summarization. BART is built on two components: a bidirectional encoder and an autoregressive decoder. Contextual information is captured from the input and the output is co-generated. A denoising objective is used for pre-training, in which corrupted text is rebuilt to its original form. The design is well-suited for providing fine-tuning capabilities in relation to specialized domains.

TABLE II
TRAINING CONFIGURATION

Parameter	Value
Base model	bart-large-cnn
Learning rate	3×10^{-5}
Scheduler	Cosine with warmup
Batch size	2 (per device)
Gradient accumulation	4
Effective batch size	8
Epochs	Up to 5
Weight decay	0.01
Label smoothing	0.1
Max input length	512
Max target length	384
Beam size	4
Length penalty	1.2

B. Training Configuration

The model was fine-tuned using the HuggingFace Transformers library together with the PyTorch framework. The main training parameters used during fine-tuning are listed in Table II.

Training was performed using mixed-precision (fp16) on an NVIDIA Tesla T4 GPU to improve computational efficiency.

C. Baseline and Model Selection

The *bart-large-cnn* model was chosen over smaller variants due to its stronger representation capacity and prior training on summarization tasks. This allows the model to better capture long-range dependencies and generate more coherent summaries after fine-tuning.

D. Key Design Considerations

Several design improvements were imposed on the model with the aim of performing better and being more stable. To ensure stable learning convergence, a cosine learning rate schedule with warmup was used during training. Label smoothing was applied to reduce overconfident predictions and enhance generalization. The target sequence was set up to 384 tokens so that the sequence does not get truncated for structured summaries, which may include several clinical fields. Beam search decoding was employed at inference stage with repetition constraints to improve diversity and quality of generations.

VII. EVALUATION METRICS

To evaluate both linguistic quality and clinical correctness, a combination of lexical, semantic, and domain-specific metrics is used.

A. ROUGE

ROUGE measures n-gram overlap between generated and reference summaries.

ROUGE-1 (unigram overlap) is computed as:

$$\text{ROUGE-1} = \frac{2 \cdot P \cdot R}{P + R}$$

ROUGE-2 measures the overlap of two-word combinations between the generated and reference summaries, helping evaluate phrase-level accuracy. **ROUGE-L** uses the longest matching sequence of words to measure how well the sentence structure is preserved.

TABLE III
EVALUATION RESULTS ON THE TEST SET

Model	R-1	R-2	R-L	BERT-F1	METEOR	NER	MEDCON
TF-IDF	0.2918	0.1751	0.1936	0.7291	0.0684	0.2459	0.5143
BART (0-shot)	0.1926	0.1169	0.1384	0.7128	0.0395	0.1814	0.4268
FT-BART	**0.4922**	**0.3381**	**0.3470**	**0.9187**	**0.2546**	**0.5818**	**0.6531**

B. BERTScore

BERTScore evaluates semantic similarity using contextual embeddings:

$$\text{BERTScore-F1} = \frac{2 \cdot P_{\text{bert}} \cdot R_{\text{bert}}}{P_{\text{bert}} + R_{\text{bert}}}$$

It captures meaning rather than exact word overlap, which is important for clinical text.

C. METEOR

METEOR evaluates summaries by matching words while also considering similar words and different word forms. It gives a more flexible evaluation than ROUGE and better matches human judgement.

D. Medical Entity Preservation

This metric measures retention of clinical entities:

$$\text{NER} = \frac{|\text{Entities}_{ref} \cap \text{Entities}_{pred}|}{|\text{Entities}_{ref}|}$$

It ensures that important medical information such as diagnoses, drugs, and procedures is preserved.

E. Clinical Concept Overlap (MEDCON)

MEDCON evaluates overlap of medical concepts:

$$\text{MEDCON} = \frac{2 \cdot P_c \cdot R_c}{P_c + R_c}$$

This metric focuses on preserving clinically meaningful concepts rather than surface text similarity.

F. Rationale

Lexical overlap scores like ROUGE and semantic similarity scores such as BERTScore are complementary. METEOR is flexible with wordings and phrases in grading. Crucial clinical data is preserved in the created summaries through the use of NER Preservation and MEDCON. These metrics give a well-rounded evaluation both in terms of summary quality and clinical value.

VIII. RESULTS AND ANALYSIS

A. Quantitative Results

Table III shows the performance of all models evaluated using different metrics on the test dataset.

The fine-tuned BART model achieves the highest performance across all metrics, clearly outperforming both extractive and zero-shot baselines.

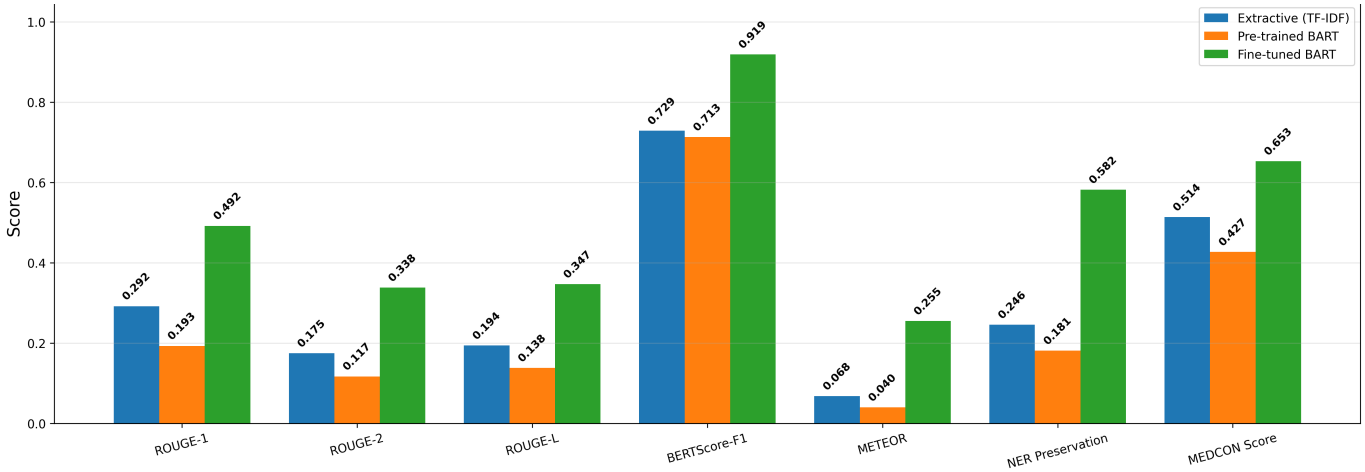


Fig. 2. All Metrics Comparison

TABLE IV
PERFORMANCE IMPROVEMENT OF FINE-TUNED BART

Metric	Gain vs TF-IDF	Gain vs 0-shot
ROUGE-1	+69%	+156%
ROUGE-2	+93%	+189%
BERTScore-F1	+26%	+29%
METEOR	+272%	+545%
NER	+137%	+221%
MEDCON	+27%	+53%

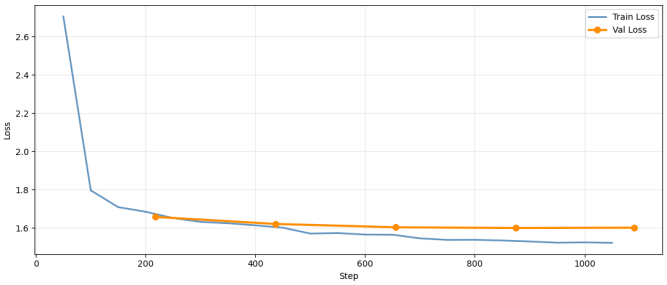


Fig. 3. Training and Validation Loss Curve for Fine-tuned BART-large-CNN

B. Performance Gains

Table IV shows the improvement of the fine-tuned model over baseline approaches.

Table IV shows the improvement achieved by the fine-tuned model compared to the baseline approaches. The improvements are observed across all evaluation metrics, with larger gains seen in METEOR and entity-level measures, indicating better semantic alignment and preservation of clinical information.

C. Training Loss Analysis

Figure 3 illustrates the training and validation loss trends during fine-tuning of the BART-large-CNN model.

The training loss gradually drops from around 2.7 to 1.52, indicating that steady decrease in the training loss was effective during fine-tuning in terms of learning and convergence. Similarly, the validation loss decreased from 1.66 and retained around 1.60 in later training steps. The training error and the validation error are very similar, indicating that the model generalises well with a low level of overfitting. The two curves are also consistent, indicating that the model is well-optimized for clinically accurate performance on the summarization task.

D. Metric-wise Analysis

ROUGE: The fine-tuned model obtained better ROUGE scores compared to the baseline models, indicating improved selection of relevant content and better generation of meaningful phrases in the summaries.

BERTScore: The achieved BERTScore value of 0.9187 shows a high semantic similarity between the generated and reference summaries, suggesting that the model is able to preserve the intended clinical meaning effectively.

METEOR: The higher METEOR score shows that the model handles different wordings and similar medical terms more effectively in clinical text.

Medical Entity Preservation: The fine-tuned model preserved more important medical information, including diseases, symptoms, medicines, and treatments, compared to the baseline models.

E. Qualitative Observations

While the fine-tuned model yields structured results, consistent with clinical data expectations like symptoms, diagnoses, and treatment plans, the zero-shot model tends to generate more coherent yet eclectic summaries for the most part, not maintaining the structured result format.

F. Discussion

On the whole, the findings demonstrate an extremely significant improvement achieved by fine-tuning within the particular

domain. Incorporating the approach, we can say that there has been more semantic alignment and more preserved entities, indicating that our proposed approach is better suited for reality-based summarization, especially when clinical applications are considered.

IX. Conclusion

In this paper, we propose a SCNS framework for fine-tuning a pre-trained BART model in an end-to-end way. The framework involves domain-specific training and evaluation using a multi-metric evaluation protocol, which evaluates the language aspect as well as the non-language aspects, such as clinical relevance.

Experimental results show that our approach performs remarkably well compared to the extractive and zero-shot baselines, in terms of all evaluation metrics considered. The results obtained by ROUGE and METEOR metrics show great performance in terms of content coverage and naturalness respectively. The excellent performance in BERTScore reflects that the language is semantic well-aligned. Moreover, the improvements in NER Preservation and MEDCON scores demonstrate better tracking of clinical concepts and named entities.

These results prove the significance of adapting to the clinical domain for clinical text summarization tasks, where better summarization results can be achieved. The produced summaries can be used directly in the clinical environment because they will be structured and more meaningful. In summary, this approach can provide a reliable way for summarizing medical text documents.

Future Work: Several directions can be used to solve identified problems. First, generalization capabilities of our model can be enhanced via training on a larger set of data and using additional clinical documents. Second, the performance can be further evaluated by employing bigger test datasets and performing expert review. Third, additional information from other documents, such as lab results and locked EHRs, can be added to achieve better context-awareness of summaries. Fourth, we can perform model compression or knowledge distillation to speed up inference process.

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